

Mathematics

Time Allowed : 2 hour _____ **Maximum Marks : 200**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. Section I contains 10 questions. Each question in this section contains 10 marks.
2. Section II contains 5 questions. Each question in this section contains 20 marks.

SECTION – I

This section contains 10 questions.

Q1: Determine the value of x satisfying the following equality:

$$|(x^4 - 4) - (x^2 + 2)| = |x^4 - 4| - |x^2 + 2|$$

Q2: Sketch the graph of $y = x \sin x$ with mathematical steps.

Q3: Evaluate the limit:

$$\lim_{x \rightarrow \frac{\pi}{4}} (\sin 2x)^{\tan^2 2x}$$

Q4: Evaluate :

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{(\sin x - \tan x)^2 + (1 - \cos 2x)^4 + x^5}{7 \tan^7 x + \sin^6 x + 2 \sin^5 x}$$

Q5: Prove that the following sequence converges and find its limit.

$$x_1 = \sqrt{2}, x_2 = \sqrt{2 + \sqrt{2}}, x_3 = \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \dots, x_n = \underbrace{\sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}}_{n \text{ radicals}}, \dots$$

Q6: For the following function, find the points of discontinuity and determine their character.

$$y = \frac{1}{u^2 + u - 2}, \text{ where } u = \frac{1}{x - 1}$$

Q7: Are the following functions identical?

a) $f(x) = \frac{x}{x}$ and $g(x) = 1$

b) $f(x) = \log(x^2)$ and $g(x) = 2 \log x$

c) $f(x) = x$ and $g(x) = (\sqrt{x})^2$

d) $f(x) = 1$ and $g(x) = \sin^2 x + \cos^2 x$

e) $f(x) = \log(x - 1) + \log(x - 2)$ and $g(x) = \log(x - 1)(x - 2)$

Q8: Test the following function for continuity and sketch its graph

$$y = \lim_{n \rightarrow \infty} \frac{x}{1 + (2 \sin x)^{2n}}$$

Q9: Will the sum of two functions $f(x) + g(x)$ be necessarily discontinuous at a point x_0 if both functions are discontinuous at $x = x_0$.

Q10: Find the constants a and b from the condition:

$$\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x + 1} - ax - b \right) = 0$$

SECTION – II

This section contains 5 questions.

Q11: Find the range of the function defined on $\mathbf{R}$ given by

$$f(x) = \begin{cases} e^{x-n} & ; n \leq x \leq n + \frac{2}{3} \\ \sin\left(\frac{3\pi}{2}x\right) & ; n + \frac{2}{3} < x < n + 1 \end{cases}$$

where $n \in \mathbf{Z}$.

Q12: A function $f : \mathbf{R} - \{-2, 2\} \rightarrow \left(-\infty, \frac{5}{2}\right] \cup (3, \infty)$ given by $f(x) = 3 + \frac{1}{|x| - 2}$. Divide the domain into minimum number of suitable subsets so that the function becomes invertible on each individual domain. Also find the inverse on each of these domains.

Q13: Prove that $[x] + [5x] + [10x] + [20x] = 36k + 35$, $k \in \mathbf{Z}$ does not have any real solution.

Q14: Prove that there exists only one continuous function $x = x(y)$ ($-\infty < y < \infty$) which satisfies the *Kepler equation* $x - \epsilon \sin x = y$ ($0 < \epsilon < 1$).

Q15: Prove that the function

$$f(x) = \begin{cases} (x+1)2^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Ascertain that on the interval $[-2, 2]$ the function takes on all intermediate values from $f(-2)$ to $f(2)$ although it is discontinuous?